Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 4 Tutorial Worksheet: STP-Based Layer 2 Attacks

Field	Detail
Based on	Week 4, STP-Based Layer 2 Attacks lecture (STP fundamentals, path cost and port selection, algorithm, versions, attacks, and six mitigations including BPDU Filter)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO2 (evaluate mitigation techniques)
Format	12 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's lecture. Several ask you to evaluate a mitigation, not just describe an attack. That's deliberate: LO2 is specifically about weighing mitigation options, not just naming them. Write full answers, not one-line definitions.

STP Fundamentals

Q1. Explain why STP is necessary in a network with redundant switch links, and what would actually happen to that network without it. *[LO1]*

STP Algorithm

Q2. Describe the three steps of the STP algorithm (root bridge election, root port election, designated port election), and explain what determines the outcome of each step. *[LO1]*

Q3. A switch has two possible paths back to the root bridge: a direct 100 Mbps link (cost 19), and an indirect path through a neighbouring switch that costs 4 to reach the root plus a further 19 to reach this switch. Calculate the Root Path Cost of each option, state which becomes this switch's Root Port, and explain your reasoning. *[LO1]*

Q4. Describe the four STP port states in order, and explain why RSTP's new port roles allow it to reach a forwarding state far faster than classic STP. *[LO1]*

STP Versions

Q5. Compare STP, RSTP, PVST+/Rapid PVST+, and MSTP. What does each version add over the one before it? *[LO1]*

STP Attacks

Q6. Describe how the STP root bridge manipulation attack works, and explain why the lack of authentication on BPDUs is what makes it possible. *[LO1]*

Q7. Explain how Topology Change (TC) Flooding/Spoofing degrades network performance. What concept from Week 3 does the resulting symptom resemble, and why? *[LO1]*

Q8. Explain how an attacker can cause a network-wide denial of service using STP without ever becoming the root bridge. *[LO1]*

STP Mitigations

Q9. For each of Root Guard, BPDU Guard, and Loop Guard, state which of this week's three attacks it would actually prevent, and explain your reasoning for each. *[LO2]*

Q10. BPDU Guard and BPDU Filter protect the same PC-facing access ports, but respond to an incoming BPDU completely differently. Describe that difference, and evaluate which one you would recommend for a corporate office's PC ports, justifying your choice. *[LO1, LO2]*

Q11. Explain the difference between what BackboneFast and UplinkFast each optimise for, and why a network might reasonably need both rather than just one. *[LO2]*

Q12. A network administrator has configured BPDU Guard on every access port, but has not touched the distribution or core switches at all. Evaluate this configuration: which attacks does it stop, and what STP-related risk remains unaddressed? *[LO2, LO3]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. This week's lab has you build a 3-tier topology, test its behaviour under default STP, then harden it (including comparing BPDU Guard against BPDU Filter directly) and test again. Configuring these mitigations yourself will cement the theory faster than rereading it.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5). The evaluation-style questions above are close in format to the structured-essay section.